

Lecture -Tutorial 1

MAT700 - Mathematical Methods for Data Mining,

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Problem 1.

Let us look at a class of 120 students and let us assume that their birthdays are independent and uniformly distributed over the 365 days of the year. We are interested in events that some students have their birthday on the same day. Show that the expectation of the number of **triples** of students having the same birthday in a class of 120 is ≈ 2.108.

Hint: K objects in N boxes. Expected number of m-tuples of objects in same box $\binom{K}{m}\frac{1}{N^{m-1}}$. Here, $K = 120$, $m = 3$, $N = 365$.

$TF \times IDF$: Reminder



Suppose we have a collection of N documents.
Let f_{ij} be the number of occurrences of term (word) i in document j.

Term Frequency

$$TF_{ij} := \frac{f_{ij}}{\max_k f_{kj}}.$$

Suppose term i appears in n_i of the N documents in the collection.

Inverse Document Frequency

$$IDF_i = \log_2 \left(\frac{N}{n_i} \right).$$

TF.IDF



Problem 2.

Suppose there is a repository of ten million documents. What(to the nearest integer) is the *IDF* for a word that appears in

(a) 40 documents,

(b) 10,000 documents?

Problem 3.

Suppose there is a repository of ten million documents, and word *w* appears in 320 of them.
In a particular document *D*, the maximum number of occurrences of a word is 15. Approximately what is the *TF*.*IDF* score for *w* if that word appears

(a) once

(b) five times?

TF.IDF



Problem 4.

Suppose there is a repository of 2048 documents, and word w appears in 128 of them. In a particular document D, the maximum number of occurrences of a word is 45. What is the $TF.IDF$ score for w in D if that word appears nine times in D?

Vector Space Model

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Problem 5.

For a given document-term matrix for a corpus of documents

	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7	Document 8
Term(s) 1	10	0	1	0	0	0	0	2
Term(s) 2	0	2	0	0	0	18	0	2
Term(s) 3	0	0	0	0	0	0	0	2
Term(s) 4	6	0	0	4	6	0	0	0
Term(s) 5	0	0	0	0	0	0	0	2
Term(s) 6	0	0	1	0	0	1	0	0
Term(s) 7	0	1	8	0	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0	0

Word vector (passage vector)

Document vector

calculate corresponding vector representation of the documents (TFxIDF matrix).

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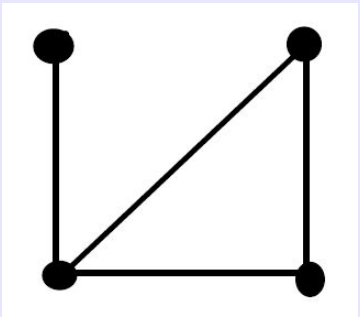
Problem 6.

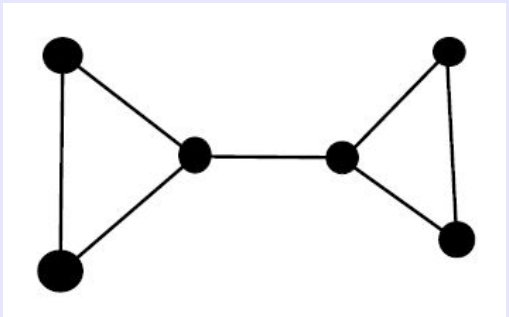
Let $G = (V, E)$ be an undirected graph . Show that

$$\sum_{v \in V} \deg(v) \;=\; 2 \cdot |E|.$$

Problem 7.

Find the adjacency matrices of the following graphs:





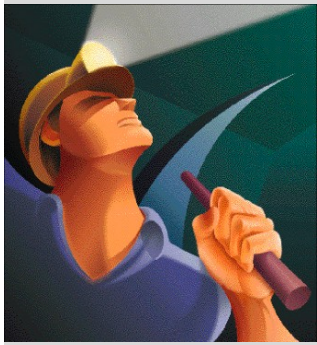
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Problem 8.

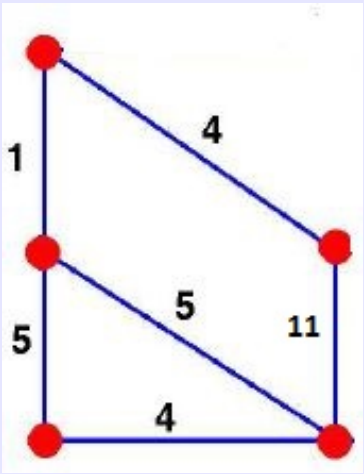
Let G be a graph with adjacency matrix A with respect to the ordering $v_1, v_2, \ldots, v_n$. Prove that the number of different walks of length r from v_i to v_j , where r is a positive integer, equals the (i, j)th entry of A^r.

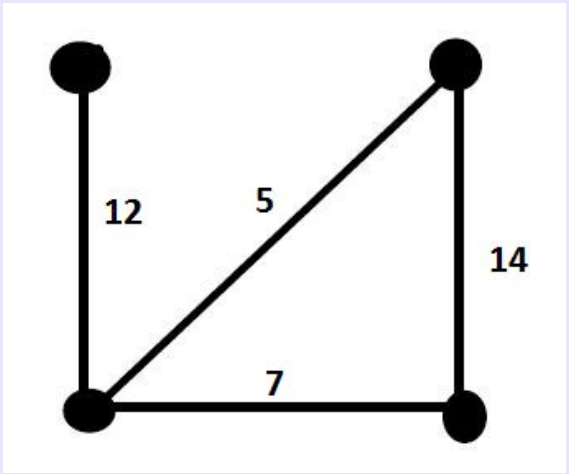
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Problem 9.

Find the adjacency matrices of the following graphs:





Jaccard Similarity



The *Jaccard similarity* of sets S and T is $|S \cap T|/|S \cup T|$, that is, the ratio of the size of the intersection of S and T to the size of their union.

Problem 10.

Compute the Jaccard similarities of each pair of the following three sets:
$\{1,2,3,4\}$, $\{2,3,5,7\}$ and $\{2,4,6\}$.

Problem 11.

Compute the Jaccard similarities of the following two sets:
$\{0,1,2,-1,-2,-3\}$ and $\{-3,3,-2,2,-1,1\}$.